

SOME NON-PRINCIPAL RIGID IDEALS IN GORENSTEIN DOMAINS OF DIMENSION ONE

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WITH AN APPENDIX BY CRAIG HUNEKE

ABSTRACT. We present an example of a rigid non-principal ideal in a one dimensional (commutative) Gorenstein domain, which contradicts a conjecture of C. Huneke and R. Wiegand. The construction and its analysis were discovered by AI, when prompted to verify the conjecture or find a counterexample. A proof of the conjecture, also discovered by AI, is presented when the ring is equicharacteristic and its embedding dimension is at most three. Another counterexample to the conjecture was discovered by Pham, also using AI. This is given in the appendix, with an independent verification that the example has the desired properties.

1. INTRODUCTION

A conjecture of Huneke and Wiegand [8, pp. 473] postulates that any ideal I in a Gorenstein (commutative, noetherian) local domain R with the property that the R -module $I \otimes_R \operatorname{Hom}_R(I, R)$ is torsion-free, must be free. The condition that the tensor product is torsion-free is equivalent to the condition that

$$\operatorname{Ext}_R^1(I, I) = 0,$$

that is to say, that the ideal I is *rigid*; see, for instance, [7, Proposition 4.3]. This conjecture has been verified in various special cases [7].

In this note we describe a general method, discovered by ChatGPT, for constructing counter-examples to this conjecture; see Propositions 2.2 and 3.2. A concrete example realizing this method is given in Example 3.3. In fact, in this example the ring R is standard-graded, and the ideal I is two-generated, by linear forms; we refer to Example 3.3 for other noteworthy features.

At about the same time as these examples were discovered, Son Pham [12] found another example, using GPT-5.6 Sol. Appendix B, due to Huneke, gives an independent verification that this example has the desired properties.

AI declaration. The constructions and results described below, including Example 3.3, and Proposition 4.2, were found by Sol Ultra on Codex, when prompted to prove the Huneke–Wiegand conjecture for Gorenstein rings or find counterexample, and separately prompted to either prove it for complete intersections or find counterexample.

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However, AI had no role to play in the writing of this manuscript, except that, once we produced a first draft, AI was used as a referee to weed out errors, mathematical and typographical.

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2. GORENSTEIN DOMAINS OF DIMENSION ONE

Given a field K and a symmetric bilinear form on a finite dimensional K -vector space V , the graded K -vector space $K \oplus V \oplus K$ has a commutative algebra structure, where the only non-trivial product $V \times V \rightarrow K$ is given by the form. This algebra is Gorenstein if and only if the form is not degenerate; see [10, (3.15)].

This section gives a method for constructing Gorenstein domains of Krull dimension one, allegedly inspired by work of Eisenbud and Popescu [5, Theorem 7.1] and Stevens [13].

Definition 2.1. Let K be a field and $K \subset L$ a finite extension such that its degree $n := [L : K]$ is even. Fix a K -subspace $H \subset L$ containing K and of index 1, that is to say, $\text{rank}_K H = n - 1$.

Let W be a K -linear subspace of H satisfying the following conditions:

$$K \subsetneq W, \quad \text{rank}_K W = \frac{n}{2}, \quad \text{and} \quad W^2 \subseteq H.$$

Let t be an indeterminate and $L[t]$ the standard graded L -algebra, so $|t| = 1$. Since $W^2 \subseteq H$ holds, the graded K -vector subspace of $L[t]$ given by

$$R := K \oplus Wt \oplus Ht^2 \bigoplus_{i \geq 3} Lt^i$$

is a K -subalgebra. Here are the principal features of this ring.

Proposition 2.2. *The ring R is graded Gorenstein (in particular, noetherian) domain of Krull dimension one. As a K -algebra it is generated in degrees one and two; if $W^2 = H$, then it is standard graded.*

Proof. Clearly, one has inclusion of subrings $K[t] \subset R \subset L[t]$. Since L/K is finite dimensional, $L[t]$ is finitely generated, and even free, as an $K[t]$ -module. Thus, its $K[t]$ -submodule R is also finitely generated, and free. This justifies the claim that the ring R is a noetherian domain and of Krull dimension one.

In the remainder of the proof, it is useful to consider the K -linear functional $f: L \rightarrow L/H \cong K$, and the symmetric K -bilinear form

$$\langle -, - \rangle: L \times L \longrightarrow K \quad \text{where } \langle x, y \rangle = f(xy).$$

This form is nondegenerate: Given $0 \neq x \in L$ one has $\langle x, x^{-1}z \rangle \neq 0$ for $z \in L \setminus H$.

For any K -subspace $U \subseteq L$ one has the orthogonal subspace

$$(2.1) \quad U^\perp := \{x \in L \mid \langle x, - \rangle = 0 \text{ on } U\}.$$

This subspace satisfies $\text{rank}_K U + \text{rank}_K U^\perp = \text{rank}_K L$. We claim

$$W^\perp = W \quad \text{and} \quad H^\perp = K.$$

Indeed, the hypothesis that $W^2 \subseteq H$ is equivalent to $W \subseteq W^\perp$; equality holds because of the hypothesis that $\text{rank}_K W = n/2$ and (2.1), which justifies the

equality on the left. As to the one on the right, again from (2.1) we deduce that $\text{rank}_K H^\perp = 1$. Since $K \subset H = \text{Ker } f$, we conclude that $H^\perp = K$.

Since $H^\perp = K$, for any element w in $W \setminus K$ one has that $wH \not\subseteq H$. Thus as a K -algebra R is generated by its elements of degree one and two; evidently, it is standard graded if also $W^2 = H$.

It remains to verify that R is Gorenstein. Since R is a domain any nonzero element, in particular t , is not a zero divisor on it. A direct calculation yields an isomorphism of K -algebras

$$R/Rt \cong K \oplus (W/K)t \oplus (H/W)t^2 \oplus (L/H)t^3.$$

The elements of degree three are in the socle. We verify that these are the only ones; since $\text{rank}_K(L/H) = 1$ this is equivalent to R/Rt Gorenstein; see [2, 3.2.10]

Suppose the residue class of wt is in the socle of R/Rt . Then, from the structure of R , we get $wH \subseteq H$, that is to say w in $H^\perp = K$. Then $wt = 0$ in R/Rt . Similarly if the residue class of ht^2 is in the socle, then $hW \subseteq H$, so h is in $W^\perp = W$, and hence $ht^2 = 0$ in R/Rt .

This completes the verification that R/Rt is Gorenstein. Since t is a homogeneous element and not a zero-divisor, we conclude that the ring R is Gorenstein; see [2, Proposition 3.1.19]. $\square$

3. RIGID IDEALS

In this section, we identify some conditions on the rings described in the previous section that ensure that they contain rigid ideals that are not principal.

In the setup of Definition 2.1, fix an element α in $W \setminus K$ and consider the ideal

$$I := (t, \alpha t)R$$

of R . This ideal is clearly not principal, and hence not free. The claim is that, for a suitable choice of subspace W , the ideal I is rigid. Not every choice of W works, as can be verified by a direct check.

We focus now on a special situation where it does, leaving the general analysis that leads to this choice to the end of this section.

Definition 3.1. Let $L = K[\alpha]$, where α has degree $n \equiv 0 \pmod{4}$, with $\alpha \notin K$, so $n \geq 4$, and $\alpha^n \in K^\times$. Thus $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ is a basis for L over K . For $i \geq 0$, let L_i denote the K -subspace of L spanned by $1, \dots, \alpha^i$. Thus one has a filtration of L by K -subspaces

$$K = L_0 \subset L_1 \subset \dots \subset L_{n-1} = L.$$

Choose a K -subspace $V \subset L$ containing $L_0 = K$ and such that the following conditions hold:

- (1) $\text{rank}_K V = n/4$;
- (2) the sum $V + \alpha V + \alpha^2 V$ is direct;
- (3) $V^2 = L_{n-4}$;
- (4) $L = VL_{n-3}$.

Conditions (1) and (4) imply $n \geq 8$. With V as above, set $W := V \oplus \alpha V$ and $H := L_{n-2}$. Recall that

$$R := K \oplus Wt \oplus Ht^2 \bigoplus_{i \geq 3} Lt^i$$

viewed as a graded K -subalgebra of $L[t]$.

Proposition 3.2. *In the setup of Definition 3.1, the ring R is a standard graded Gorenstein domain of Krull dimension one. The ideal $(t, \alpha t)$ is rigid, but not free.*

Proof. The claims about the ring R follows from Proposition 2.2; observe that $W^2 = H$, by condition (3) in Definition 3.1.

To verify that $(t, \alpha t)$ is rigid, we use the criterion [7, Corollary 4.9]; that statement is for local rings, but the argument applies equally well for homogeneous ideals in graded rings. Thus, we have to verify the equality:

$$((t) :_R \alpha t) \cap ((\alpha t) :_R t) = ((t) :_R \alpha t)((\alpha t) :_R t).$$

To begin with, observe that

$$((\alpha t) :_R t) = \alpha \cdot ((t) :_R \alpha t);$$

keep in mind that α itself is not an element of R . Thus the desired equality is that

$$J \cap \alpha J = \alpha J^2 \quad \text{where } J = ((t) :_R \alpha t).$$

Since $(t) = Kt \oplus Wt^2 \oplus Ht^3 \oplus \bigoplus_{i \geq 4} Lt^i$, using condition (2) in Definition 3.1 a direct verification yields

$$J = Vt \oplus L_{n-3}t^2 \bigoplus_{i \geq 3} Lt^i \quad \text{and hence} \quad \alpha J = \alpha Vt \oplus \alpha L_{n-3}t^2 \bigoplus_{i \geq 3} Lt^i.$$

Observing that $V \cap \alpha V = 0$, by condition (2) in Definition 3.1, one thus gets that

$$J \cap \alpha J = (L_{n-3} \cap \alpha L_{n-3})t^2 \bigoplus_{i \geq 3} Lt^i.$$

On the other hand, a direct computation yields the first equality below

$$\begin{aligned} \alpha J^2 &= \alpha(V^2t^2 \oplus VL_{n-3}t^3 \bigoplus_{i \geq 4} Lt^i) \\ &= \alpha(L_{n-4}t^2 \oplus Lt^3 \bigoplus_{i \geq 4} Lt^i) \\ &= \alpha L_{n-4}t^2 \oplus \bigoplus_{i \geq 3} Lt^i. \end{aligned}$$

The second equality holds by conditions (3) and (4) in Definition 3.1, and the third follows from $\alpha L = L$. Thus, to verify that $J \cap \alpha J = \alpha J^2$, it remains to verify that

$$L_{n-3} \cap \alpha L_{n-3} = \alpha L_{n-4}.$$

This is clear, given the description of the L_i . □

A numerical constraint. In the notation above, suppose $d := \text{rank}_K V$, so that $\text{rank}_K W = 2d$ and $\text{rank}_K L = 4d$. Thus the condition that $V^2 = L_{n-4}$, from Definition 3.1(2), implies that

$$\binom{d+1}{2} = \text{rank}_K \text{Sym}_K^2(V) \geq \text{rank}_K L_{n-4} = 4d - 3.$$

This leads to the condition that

$$(d-1)(d-6) \geq 0.$$

Since $n \geq 8$, the smallest possible value for d that would permit V to satisfy the constraints in Definition 3.1 is 6. Then $\text{rank}_K L = 24$.

When $d = 6$, one has

$$\operatorname{rank}_K \operatorname{Sym}_K^2(V) = 21 = \operatorname{rank}_K L_{20}$$

so condition (2) in Definition 3.1 is equivalent to the non-vanishing of a determinant of size 21×21 . These considerations motivate the following example.

Example 3.3. Let $K = \mathbb{Q}$ and $L := \mathbb{Q}[\alpha]/(\alpha^{24} - 2)$. The polynomial $x^{24} - 2$ is irreducible, by, say the Eisenstein criterion, so $\operatorname{rank}_K L = 24$. Let V be the $\mathbb{Q}$ -subspace of L spanned by the six elements

$$\begin{aligned} &1, \quad \alpha - \alpha^{15}, \quad \alpha^2 - \alpha^{12} - \alpha^{16} \\ &\alpha^4 - \alpha^{14}, \quad \alpha^6 - \alpha^{13} + \alpha^{20}, \quad \alpha^9 + \alpha^{16} + \alpha^{19}. \end{aligned}$$

One can verify directly by hand, or using a computer algebra system, that V satisfies the conditions listed in Definition 3.1; see Appendix A for a *Macaulay2* verification. Thus Propositions 2.2 and 3.2 justify the following claims.

With V and L as above, and $W := V \oplus \alpha V$, the subring

$$R := \mathbb{Q} \oplus Wt \oplus L_{22}t^2 \oplus_{i \geq 3} Lt^i$$

of $L[t]$ is a standard graded Gorenstein $\mathbb{Q}$ -algebra and a domain. Moreover, the ideal $I := (t, \alpha t)$ is rigid, but not principal.

The ideal I has the added property that the evaluation map

$$I \otimes_R \operatorname{Hom}_R(I, R) \longrightarrow R$$

is an isomorphism onto $\mathfrak{m}$, the homogeneous maximal ideal of R .

Remark 3.4. One can localize at the homogeneous maximal ideal $\mathfrak{m} := R^{\geq 1}$ and get an example of local Gorenstein domain of Krull dimension one that provides a counterexample to the conjecture of Huneke and Wiegand.

A commutative algebra analogue of a conjecture of Auslander and Reiten [1] states that over any Gorenstein ring R , if a finitely generated R -module M satisfies $\operatorname{Ext}_R^i(M, M) = 0$ for $i \geq 1$, then M is free. For the ring R and ideal I in Example 3.3 one can verify that

$$\operatorname{Ext}_R^2(I, I) \neq 0.$$

Thus I is not a counterexample to the Auslander-Reiten conjecture.

An ansatz leading to Definition 3.1. Let R be a ring as in Definition 2.1 and set $I = (t, \alpha t)$, for some element α in $L \setminus K$. As explained in the proof of Proposition 3.2, the ideal I is rigid precisely when there is an equality

$$(3.1) \quad J \cap \alpha J = \alpha J^2, \quad \text{where } J := ((t) :_R \alpha t).$$

Consider the following K -subspaces

$$V := \alpha^{-1}W \cap W \quad \text{and} \quad E := \alpha^{-1}H \cap H.$$

It is straightforward to verify that

$$J = Vt \oplus Et^2 \oplus_{i \geq 3} Lt^i,$$

and hence condition (3.1) translates to

$$(3.2) \quad V \cap \alpha V = (0), \quad E \cap \alpha E = \alpha V^2, \quad \text{and} \quad L = VE.$$

Keep in mind that for any K -subspace U of L , one has $L = \alpha U$ if and only if $L = U$. The first condition means that $V \oplus \alpha V \subseteq W$ so a first guess is to take

$$W := V \oplus \alpha V.$$

Since $\text{rank}_K W$ should be $n/2$ for Proposition 2.2 to apply, $\text{rank}_K V = n/4$ must hold, which explains condition (1) in Definition 3.1.

Then since $V = \alpha^{-1}W \cap W$, it follows that the sum

$$V + \alpha V + \alpha^2 V$$

is direct, which explains condition (2) in Definition 3.1.

Now suppose that $L = K[\alpha]$ is as in Definition 3.1. In the notation introduced there, one has $H = L_{n-2}$, so

$$E = \alpha^{-1}H \cap H = L_{n-3}.$$

Thus the two conditions on the right in (3.2) translate to conditions (3) and (4) in Definition 3.1.

4. CODIMENSION TWO

In this section we verify that Huneke–Wiegand conjecture holds when, in addition, the Gorenstein ring R is equicharacteristic of codimension at most 2; see Proposition 4.2 below. A key input in the argument is a result of Neubauer and Saltman [11]. We begin by recasting it form that is convenient for our application.

Proposition 4.1. *Let k be a field and A a finitely dimensional commutative k -algebra. Assume the k -algebra A is two-generated. For any finitely generated faithful A -module M , there are inequalities*

$$\text{rank}_k A \leq \text{rank}_k M \leq \text{rank}_k \text{End}_A(M).$$

Moreover, if either inequality is an equality, then both become equalities.

There is an analogue of this result, due to Schur and Jacobson, for arbitrary A but that involves a term that is quadratic in $\text{rank}_k M$; see [4] for an elegant proof of this result. However, this does not seem to help for applications to the conjecture of Huneke and Wiegand.

Proof. The hypothesis that A is two-generated and M is a faithful A -module is tantamount to saying that there are commuting k -linear operators $x, y: M \rightarrow M$ such that $A \cong k[x, y]$, the k -subalgebra of $\text{End}_k(M)$ generated by x, y . Since $\text{End}_A(M)$ is the k -subalgebra of $\text{End}_k(M)$ consisting of operators that commute with both x and y , it remains to apply [11, Theorem 1.1, Lemmas 1.2 & 1.3]. $\square$

Here is the avowed result about rigid ideals. We write $\text{emb dim } A$ to denote the embedding dimension of a local ring A .

Proposition 4.2. *Let $(R, \mathfrak{m}, k)$ be a Gorenstein local domain of Krull dimension one. Assume $\text{emb dim } R \leq 3$ and that there exists an element u in $\mathfrak{m} \setminus \mathfrak{m}^2$ such that the artinian ring R/Ru contains k .*

Every rigid ideal in R is principal; equivalently, free.

Proof. The result is well-known when $\text{emb dim } R \leq 2$; see [8, Theorem 3.1]. In the rest of the proof we assume $\text{emb dim } R = 3$.

Let I be a rigid ideal in R . Set $S := R/Ru$ and $M := I/Iu$. Our hypothesis on means that S is a local k -algebra, of embedding dimension two.

To verify I is free, it suffices to verify that the S -module M is free. To that end, since the ring S is Gorenstein, it suffices to verify that M is a faithful S -module, with $\text{rank}_k S = \text{rank}_S M$.

Applying $\text{Hom}_R(I, -)$ to the exact sequence of R -modules

$$0 \longrightarrow I \xrightarrow{u} I \longrightarrow M \longrightarrow 0$$

yields the exact sequence of R -modules

$$0 \longrightarrow \text{End}_R(I) \xrightarrow{u} \text{End}_R(I) \longrightarrow \text{Hom}_R(I, M) \longrightarrow \text{Ext}_R^1(I, I) = 0$$

The equality holds because I is rigid. Observing that $\text{Hom}_R(I, M) \cong \text{End}_S(M)$, one deduces from the sequence above that

$$\text{End}_S(M) \cong \frac{\text{End}_R(I)}{\text{End}_R(I)u}$$

as S -modules. This justifies the first equality below

$$(4.1) \quad \text{rank}_k \text{End}_S(M) = \text{rank}_k \left(\frac{\text{End}_R(I)}{\text{End}_R(I)u} \right) = \text{rank}_k S = \text{rank}_k M.$$

The second and third equalities holds because $\text{End}_R(I)$ and I are both R -modules of rank one, and R is a domain of Krull dimension one; see, for instance, [2].

Let $\bar{S}$ be the image of S in $\text{End}_S(M)$ under the natural map $S \rightarrow \text{End}_S(M)$. Then $\bar{S}$ is also a commutative local k -algebra, generated by two elements. Thus Proposition 4.1 yields inequalities

$$\text{rank}_k \bar{S} \leq \text{rank}_k M \leq \text{rank}_k \text{End}_S(M),$$

and also that if either inequality is an equality, then both are. Given (4.1) it follows that there are equalities

$$\text{rank}_k S = \text{rank}_k \bar{S} = \text{rank}_k \text{End}_S(M).$$

We deduce that the natural map is an isomorphism: $S \xrightarrow{\cong} \text{End}_S(M)$, in particular, that M is a faithful S -module. $\square$

Remark 4.3. A Gorenstein ring R satisfying $\text{emb dim } R - \dim R \leq 2$ is complete intersection. Thus the ring R in Proposition 4.2 is a complete intersection. It remains an open question whether the conjecture of Huneke and Wiegand holds when R is, in addition, a complete intersection.

APPENDIX A. VERIFICATION WITH MACAULAY2

Here we share the simple *Macaulay2* script that we used to verify that the subspace V in Example 3.3 satisfies the conditions in Definition 3.1; it is available from [3].

```
-- Setting up K and L

K = QQ; L = K[a]/ideal(a^24-2);

-- Defining a function to compute the standard coordinates
-- of a polynomial in L

polyToList = p -> ( apply(24, i -> coefficient(a^i,p)) );

-- Setting up the subspace V of L
-- and verifying condition 3.1(1)

gensV = {1_L, a - a^15, a^2 - a^12 - a^16, a^4 - a^14,
         a^6 - a^13 + a^20, a^9 + a^16 + a^19};
```

```

V = image transpose matrix apply( gensV, b -> polyToList(b) );
rank V -- ourput: 6 cf. 3.1(1)

-- Setting up alpha times V and alpha^2 times V

gensaV = apply( gensV, b-> a*b );
aV = image transpose matrix apply( gensaV, b -> polyToList(b) );
rank aV -- output: 6

gensa2V = apply( gensV, b-> a^2*b );
a2V = image transpose matrix apply( gensa2V, b -> polyToList(b) );
rank a2V -- output: 6

-- Verifying condition 3.1(2)

rank(V+aV+a2V) == rank V + rank aV + rank a2V -- output: true

-- Verifying condition 3.1(3)

gensV2 = flatten apply( 6,i -> apply( i+1, j -> (gensV#i)*(gensV#j) ));
V2 = image transpose matrix apply( gensV2, b -> polyToList(b) );
rank V2 -- output: 21

basisL20 = apply( 21, i -> a^i );
L20 = image transpose matrix apply( basisL20, b -> polyToList(b) );
V2 == L20 -- output: true

-- Verifying condition 3.1(4)

gensVL21 = flatten apply( gensV, b -> apply( 22, i -> a^i*b ));
VL21 = image transpose matrix apply( gensVL21, b -> polyToList(b) );
rank VL21 -- output: 24, so it is all of L

```

One can realize the ring from Example 3.3 as a quotient of a polynomial algebra. Set $Q = \mathbb{Q}[x_1, \dots, x_6, y_1, \dots, y_6]$ and consider the canonical ring homomorphism $\phi: Q \rightarrow L[t]$ that maps the indeterminates to the basis for $Wt = (V \oplus \alpha V)t$ so that

$$\begin{aligned}
x_1 &\mapsto t, \dots, x_6 \mapsto (\alpha^9 + \alpha^{16} + \alpha^{19}) \\
y_1 &\mapsto \alpha t, \dots, y_6 \mapsto \alpha(\alpha^9 + \alpha^{16} + \alpha^{19})t.
\end{aligned}$$

By construction, the image of ϕ in degree 1 is R_1 , and since R and Q are standard graded, the kernel of ϕ is the defining ideal of R as a quotient of Q . Recalling that the Hilbert series of R is $1+12t+23t^2+\sum_{i=3}^{\infty} 24t^i$ and $\text{rank}_K \text{Sym}^2(Q_1) = \binom{13}{2} = 78$, one sees that the kernel of ϕ is generated by $78 - 23 = 55$ quartics. Some quite conspicuous, like $x_1y_2 - x_2y_1, \dots, x_1y_6 - x_6y_1$, and others less so; they are all listed in [3] along with code that verifies this realization of the Gorenstein ring R .

APPENDIX B. A SEMIGROUP EXAMPLE BY CRAIG HUNEKE

Son Pham's preprint [12] gives another candidate counterexample to the Huneke–Wiegand conjecture. The example is a numerical semi-group ring. In this section, we give an independent check of Pham's discovery. It uses only the definitions of

the semi-group and the relevant ideal in the semi-group ring, which are recalled further below, but none of the computations or results from [12].

In what follows $[u, v]$ denotes all integers between integers u and v . Throughout Γ is the numerical semi-group with 26 generators:

$$(B.1) \quad \Gamma = \langle [56, 58] \cup \{63, 64\} \cup [70, 83] \cup \{87, 89, 90, 93\} \cup [95, 97] \rangle.$$

Let k be a field, $A := k[\Gamma]$ the corresponding semi-group ring, and consider

$$I = (a, b) \quad \text{where } a = t^{56} \text{ and } b = t^{70},$$

an ideal in A . Set $\mathfrak{m} = A^+$ and $R := A_{\mathfrak{m}}$, the localization of A at its irrelevant ideal. Here is the sketch of the argument verifying that the local ring R and ideal IR give the desired counter-example.

A semi-group computation proves that Γ is symmetric, so Kunz's theorem [9], see also [2, Theorem 4.4.8], yields that R is Gorenstein. Independently, the calculation of $((a) : \mathfrak{m})$ proves that the parameter ideal (a) is irreducible and hence gives a second proof that R is Gorenstein. Direct membership tests in Γ give

$$((a) : b) \cap ((b) : a) = ((a) : b)((b) : a).$$

Results of Huneke–Iyengar–Wiegand [7] then show that I is rigid and $I \otimes_R I^*$ is torsionfree. Since $14 \notin \Gamma$ the ideal I is not principal, independently confirming the candidate counterexample found by Pham.

An independent membership certificate for Γ . Let G be the generators of the numerical semi-group in (B.1). For $C \subseteq \Gamma$ write

$$(t^C) := A(t^c : c \in C).$$

The minimal monomial generators of a monomial ideal in A are the exponents minimal for

$$x \preceq y \iff y - x \in \Gamma.$$

All monomial calculations are first made in A . Localization commutes with the finitely generated ideal quotients used below, so the same displayed monomials generate the corresponding ideals in R .

Below the integer $112 = 2 \cdot 56$ the nonzero elements of Γ are exactly the elements of G . Direct addition gives

$$(B.2) \quad (G + G) \cap [0, 181] = [112, 116] \cup [119, 122] \cup [126, 180].$$

Every sum of three positive elements of Γ that is at most 181 lies in $[168, 180]$, already contained in (B.2); a sum of four positive elements is at least 224. Hence

$$(B.3) \quad \begin{aligned} \Gamma \cap [0, 181] &= \{0\} \cup [56, 58] \cup \{63, 64\} \cup [70, 83] \\ &\quad \cup \{87\} \cup \{89, 90\} \cup \{93\} \cup [95, 97] \\ &\quad \cup [112, 116] \cup [119, 122] \cup [126, 180]. \end{aligned}$$

Adding 56 to $[126, 180]$ gives $[182, 236] \subset \Gamma$ and $237 = 180 + 57 \in \Gamma$. Repeated addition of 56 proves $[182, \infty) \subseteq \Gamma$. Since (B.3) shows $181 \notin \Gamma$, one gets that the conductor and Frobenius number of Γ are

$$c(\Gamma) = 182 \quad \text{and} \quad F(\Gamma) = 181,$$

respectively. The gaps through 181 are

$$(B.4) \quad \begin{aligned} &[1, 55] \cup [59, 62] \cup [65, 69] \cup [84, 86] \cup \{88\} \cup \{91, 92\} \\ &\cup \{94\} \cup [98, 111] \cup \{117, 118\} \cup [123, 125] \cup \{181\}. \end{aligned}$$

Comparison of (B.3) and (B.4) gives for every $0 \leq n \leq 181$

$$n \in \Gamma \iff 181 - n \notin \Gamma.$$

Thus Γ is symmetric. Kunz's theorem [9] says that a numerical semi-group ring is Gorenstein exactly when its semi-group is symmetric. Therefore R is a one-dimensional Gorenstein local domain.

The two principal colon ideals. For a monomial t^x in A one has

$$t^x \in ((a) :_A b) \iff x \text{ and } x + 14 \text{ are in } \Gamma.$$

Set $E = \{x \in \Gamma \mid x + 14 \in \Gamma\}$. Directly using (B.3) and the conductor gives

$$(B.5) \quad E = [56, 58] \cup [63, 64] \cup \{73\} \cup \{75, 76\} \cup \{79\} \cup [81, 83] \\ \cup [112, 116] \cup [119, 122] \cup [126, 166] \cup [168, 180] \cup [182, \infty).$$

Let $B = \{56, 57, 58, 63, 64, 73, 75, 76, 79, 81, 82, 83\}$. Direct addition gives

$$(B.6) \quad B + G = [112, 116] \cup [119, 122] \cup [126, 166] \cup [168, 180].$$

Also one has

$$[182, 222] = 56 + [126, 166], \quad 223 = 57 + 166, \\ [224, 236] = 56 + [168, 180], \quad 237 = 57 + 180, .$$

Thus $B + \Gamma$ contains $[182, 237]$, and repeated addition of 56 covers all larger integers. Comparing (B.5) and (B.6) proves $E = B + \Gamma$. Since the width of B is $27 < 56$, its elements are pairwise incomparable for $\preccurlyeq_\Gamma$. Hence

$$((a) :_R b) = (t^{56}, t^{57}, t^{58}, t^{63}, t^{64}, t^{73}, t^{75}, t^{76}, t^{79}, t^{81}, t^{82}, t^{83})R.$$

Likewise

$$t^x \in ((b) :_R a) \iff x \text{ and } x - 14 \text{ are in } \Gamma.$$

Writing $u = x - 14$ shows that this exponent set is $14 + E$ and therefore

$$((b) :_R a) = (t^{70}, t^{71}, t^{72}, t^{77}, t^{78}, t^{87}, t^{89}, t^{90}, t^{93}, t^{95}, t^{96}, t^{97})R.$$

Intersection and product. Intersecting E with $14 + E$ gives

$$(B.7) \quad E \cap (14 + E) = [126, 130] \cup [133, 136] \cup [140, 166] \\ \cup [168, 180] \cup [182, 194] \cup [196, \infty).$$

One has $B + B = [112, 116] \cup [119, 122] \cup [126, 152] \cup [154, 166]$. The product $((a) :_R b)((b) :_R a)$ is generated in exponents by

$$B + (14 + B) = 14 + (B + B) = X$$

where $X = [126, 130] \cup [133, 136] \cup [140, 166] \cup [168, 180]$. The width of X is $54 < 56$, so all 49 elements of X are minimal.

The product is automatically contained in the intersection. Conversely, direct addition gives

$$X + G = [182, 194] \cup [196, 277].$$

Since the product is closed under addition by 56, the interval $[196, 277]$ propagates to all of $[196, \infty)$. Comparing with (B.7) justifies the equality

$$(B.8) \quad ((a) :_R b) \cap ((b) :_R a) = ((a) :_R b)((b) :_R a) = (t^x : x \in X)R.$$

The socle colon and the Gorenstein property, again. A gap $q \notin \Gamma$ is pseudo-Frobenius if

$$q + \gamma \in \Gamma \quad \text{for every } \gamma \in \Gamma \setminus \{0\}.$$

If $q < 181$ is a gap, symmetry makes $181 - q$ a positive element of Γ , so q is not pseudo-Frobenius because $q + (181 - q) = 181 \notin \Gamma$. Conversely, $181 + g \geq 237 \in \Gamma$ for every $g \in G$, and closure gives the condition for every positive $\gamma \in \Gamma$. Thus the pseudo-Frobenius number of Γ is

$$(B.9) \quad \text{PF}(\Gamma) = \{181\}.$$

Now $t^x \in ((a) :_A \mathfrak{m})$ exactly when

$$x \in \Gamma \quad \text{and} \quad x + g - 56 \in \Gamma \quad \text{for every } g \in G.$$

Set $q = x - 56$. If $q \in \Gamma$, then $t^x \in (a)$. If $q \notin \Gamma$, the colon condition makes q pseudo-Frobenius. The only negative possibility is $x = 0$, which fails for $g = 57$. Equation (B.9) therefore gives the single additional exponent $56 + 181 = 237$. Since $181 \notin \Gamma$, this generator is not in (t^{56}) , and

$$(B.10) \quad ((a) :_R \mathfrak{m}) = (t^{56}, t^{237})R.$$

The socle of the artinian local ring $R/(a)$ is

$$\text{Soc}(R/(a)) = (0 :_{R/(a)} \mathfrak{m}) = \frac{((a) :_R \mathfrak{m})}{(a)} \cong \overline{Rt^{237}} \cong R/\mathfrak{m} \cong k,$$

where the first isomorphism is from (B.10). Indeed, $\overline{t^{237}} \neq 0$ because $237 - 56 = 181 \notin \Gamma$ and the class is killed by $\mathfrak{m}$ by definition of the colon. Since $R/(a)$ is artinian with one-dimensional socle, it is Gorenstein. This gives a second proof, independent of Kunz's theorem, that R is Gorenstein; see [2, Proposition 3.1.19].

The counterexample. Here then is the independent verification that Pham's example has the desired properties.

Theorem B.1. *With A as above and $R = A_{\mathfrak{m}}$, the ideal $I = (t^{56}, t^{70})R$ is non-free, torsion-free, and of rank one, while the R -module $I \otimes_R \text{Hom}_R(I, R)$ is torsion-free.*

Proof. Clearly R is a one-dimensional local domain, and we have already verified, in two different ways, that it is Gorenstein. Both t^{56} and t^{70} are nonzero and hence regular. So (B.8) and [7, Corollary 5.7] yield that I is rigid. Since I is a torsion-free module of rank one, [7, Proposition 5.1] makes $I \otimes_R \text{Hom}_R(I, R)$ torsion-free.

Finally, a rank-one ideal in a local domain is free if and only if it is principal. Suppose $I = (f)$. Its least valuation forces $\nu(f) = 56$. Since $t^{56} = fr$ for some $r \in R$, one has $\nu(r) = 0$, so r is a unit. $\square$

Remark B.2 (Fresh mechanical audit). After the proofs were written, every numerical assertion in this note was recomputed from scratch using only the list G from (B.1). The check did not read any earlier calculation or any file from Pham's preprint. It used Dijkstra's shortest-path algorithm on the residue graph $\mathbb{Z}/56\mathbb{Z}$ to compute the Apéry vector

$$w_r := \min\{n \in \Gamma \mid n \equiv r \pmod{56}\}.$$

Thus $n \in \Gamma$ was tested by the exact criterion $n \geq w_{n \bmod 56}$. Each monomial ideal J was independently encoded by its 56-entry vector

$$w_r(J) := \min\{n \mid t^n \in J, n \equiv r \pmod{56}\}.$$

Minimal generators were reduced with respect to $\preceq_\Gamma$; intersections were taken coordinate-wise; and the product vector was obtained by all pair sums followed by the Γ -closure. The two 56-entry vectors for the intersection and product were identical. The same audit checked every finite interval identity displayed in this note, $\text{PF}(\Gamma) = \{181\}$, and the single surviving monomial class in $((a) : \mathfrak{m})/(a)$.

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